Appendix

Syllabus 2a Mathematics Test

Below the minimum of expected knowledge for mathematics is presented. Note that the questions on the respective test might consist of a combination of multiple topics. The content in this syllabus is based on the material covered in Dutch VWO (i.e. pre-university education) schools.
The standard mathematical terms are written in **boldface**. Note that these terms might be very different in your native language. It is advised to check those terms carefully, look up the terms that you do not recognize and make a list of translations to your native language.

Mathematics

The math problems can and have to be solved exactly, i.e. without using approximation techniques or a calculator. Moreover, unless stated otherwise, this also implies that you should not round your answers (e.g. 0.33 is not considered the same as 1/3).

1. Functions and Graphs

   i The candidate is able to recognize and construct **compositions** of standard **functions**. Standard functions include.

   - **polynomial functions**,
   - ***n*-root functions** ($\sqrt[n]{x}$, $x^{\frac{1}{n}}$),
   - **power functions** (x^a, a fixed),
   - **exponential functions** (a^x, a fixed. Specifically e^x),
   - **logarithms** ($\log_a(x)$, a fixed. Specifically the **natural logarithm** $\ln(x)$),
   - **trigonometric functions** ($\sin(x)$, $\cos(x)$ and $\tan(x)$),
   - the **absolute value function** ($|x|$).

   ii The candidate is able to analyze, and transform (compositions of) these standard functions, to determine **limits**, **domain**, **range**, **asymptotes** and **symmetry**-points or -lines and to draw and recognize graphs of (compositions of) these functions.

   iii The candidate understands the concept of **inverse functions**, and can find the inverse of (compositions of) standard functions.

2. Algebraic manipulations an solving equations

   i The candidate can rewrite expressions to isolate a variable and can substitute expressions into a given function.

   ii The candidate is able to rewrite expressions into simplified form and use this skill to manipulate and solve **equations** and **inequalities** of the form $f(x) = g(x)$, $f(x) \leq g(x)$, $f(x) \geq g(x)$, $f(x) < g(x)$, $f(x) > g(x)$ and $f(x) \neq g(x)$, where f and g are (compositions of) standard functions (see 1i)

iii The candidate is able to find **roots of a function** ($f(x) = 0$) using **factorization techniques**. The candidate is able to use the **quadratic formula** to find roots of **quadratic polynomials** ($ax^2 + bx + c = 0$).

iv The candidate can solve **systems of linear equations**, $\begin{cases} ax + by = c, \\ dx + ey = f, \end{cases}$ with a, b, c, d, e, f constants.

3. Differential Calculus

   i The candidate knows the **derivatives** of standard functions, and is able to apply the **sum rule**, **product rule**, **quotient rule**, and **chain rule** to determine derivatives of functions composed of standard functions.

   ii The candidate is able to determine the first derivative ($f'(x), \frac{dy}{dx}, \frac{d}{dx}f(x)$) and second derivative ($f''(x), \frac{d^2y}{dx^2}, \frac{d^2}{dx^2}f(x)$) of functions and to use these to determine **locally increasing** and **locally decreasing** behavior, **extremal values**, and **inflection points**.

   iii The candidate is able to apply differentiation to determine the **slope of a graph** and the local **tangent lines** and **normal lines** to the graph of a function, to construct and solve a optimization problems, and to solve problems concerning **distance**, **velocity** and **acceleration**.

4. Integral Calculus

   i The candidate understands the concept of **integration** and related terms (including **limits of integration**, **definite/indefinite integrals** and the **integration constant**).

   ii The candidate is able to determine **antiderivatives** (also called **primitive functions**) of standard functions, and is able to use this to calculate definite and indefinite integrals of functions of the form $cf(ax + b) + d$, with a, b, c, d constants and f a standard function.

   iii The candidate is able to apply integration to determine **surface area** and **volume** of a **solid of revolution** and the **mean value** of a function.

5. Trigonometry

   i The candidate understands the trigonometric functions $\sin(x)$, $\cos(x)$ and $\tan(x)$ and their relation to the **unit circle**. The candidate understands the terms **amplitude**, **phase**, **period**, and **frequency** and can relate those to the parameters in a **sinusoidal function** such as $f(t) = d + a\sin(b(t - c))$. The candidate is able to convert **degrees** to **radians** and vice-versa.

   ii The candidate knows the exact values of $\sin(\theta)$, $\cos(\theta)$ and $\tan(\theta)$ for the following angles (in radians) $\theta = 0, \frac{1}{6}\pi, \frac{1}{4}\pi, \frac{1}{3}\pi$ or $\frac{1}{2}\pi$, as well as **integer** multiples of these angles.

   iii The candidate knows is able to use periodicity and symmetry properties of $\sin(\theta)$, $\cos(\theta)$ and $\tan(\theta)$.

   iv The candidate is able to find all solutions of equations $\sin(x) = c$, $\cos(x) = c$ and $\tan(x) = c$, and of $\sin(f(x)) = \sin(g(x))$, $\cos(f(x)) = \cos(g(x))$ and $\tan(f(x)) = \tan(g(x))$, where c is a constant and $f(x)$ and $g(x)$ are **linear functions** of x.

   v The candidate is able to find all solutions of equations $\sin(x) = c$, $\cos(x) = c$ and $\tan(x) = c$, and of $\sin(f(x)) = \sin(g(x))$, $\cos(f(x)) = \cos(g(x))$ and $\tan(f(x)) = \tan(g(x))$, where c is a constant and $f(x)$ and $g(x)$ are linear functions of x.

vi The candidate is able to solve inequalities $\sin(f(x)) \leq c$, $\cos(f(x)) \leq c$ and $\tan(f(x)) \leq c$, where c is a constant and $f(x)$ and $g(x)$ are linear functions of x. The same for $\leq$ replaced with $<$, $>$ or $\geq$.

vii The candidate is able to apply the **Pythagorean identity** $\sin^2(x) + \cos^2(x) = 1$, **sum and difference identities** and **double angle formulae**.

6. Geometry

   i The candidate is able to determine the **surface area** and **perimeter length** of two-dimensional shapes including **triangles**, **rectangles**, **circles**, etc. The candidate is able to determine the volume and surface area of three-dimensional objects including **cubes**, **pyramids**, **cylinders**, **cones**, etc.

   ii The candidate can use properties of lines, triangles, circles, and **quadrilaterals** to determine **lengths** and **angles**. The candidate knows and can use the properties of a **right-triangle**, **isosceles triangle**, and **equilateral triangle**.

   iii The candidate can use the **Pythagorean theorem**, relations between sin, cos and tan, the **law of sines** and the **law of cosines** to determine lengths and angles in triangles.

   iv The candidate can formulate equations for lines and circles, and knows the relations between the slopes of normal and tangent lines.

   v The candidate is able to find the **points of intersection** between lines and circles.

7. Vectors

   i The candidate understands the concept of a **vector**, and can determine the **norm** (i.e. length) and **direction** of a vector.

   ii The candidate can **decompose** vectors into **components**, can multiply a vector with a **scalar**, and can add and subtract vectors. The candidate can calculate the **dot product** of two vectors, and can use it for the calculation of angles and distances and to detect **orthogonality**.

   iii The candidate can calculate **speed**, velocity and acceleration of a moving point whose path is described by a time-dependent vector representation.

   Remark:
   Vectors will be denoted boldface or with an arrow: $\mathbf{v}$ or $\vec{v}$. When expressed in components, a vector will denoted using round brackets, e.g. $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$. The norm (= length) of a vector $\mathbf{v}$ will be denoted as $\|\mathbf{v}\|$.

Formula sheet: Mathematics - v2019.1

Trigonometry

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1 \tag{1}$$

Angle sum and difference identities

$$\cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta) \tag{2}$$

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) \tag{3}$$

$$\sin(\alpha - \beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta) \tag{4}$$

$$\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta) \tag{5}$$

$$\tan(\alpha - \beta) = \frac{\tan(\alpha) - \tan(\beta)}{1 + \tan(\alpha)\tan(\beta)} \tag{6}$$

$$\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)} \tag{7}$$

Double-angle formulae

$$\cos(2x) = \cos^2(x) - \sin^2(x) \tag{8}$$

$$= 2\cos^2(x) - 1 \tag{9}$$

$$= 1 - 2\sin^2(x) \tag{10}$$

$$\sin(2x) = 2\sin(x)\cos(x) \tag{11}$$

$$\tan(2x) = \frac{2\tan(x)}{1 - \tan^2(x)} \tag{12}$$

Integrals

$$\int x^a \ \mathrm{d}x = \frac{x^{a+1}}{a+1} + C \qquad (a \neq -1) \tag{13}$$

$$\int a^x \ \mathrm{d}x = \frac{a^x}{\ln(a)} + C \qquad (a \neq 1) \tag{14}$$

$$\int \frac{1}{x} \ \mathrm{d}x = \ln|x| + C \tag{15}$$

$$\int e^x \ \mathrm{d}x = e^x + C \tag{16}$$

$$\int \ln(x) \ \mathrm{d}x = x\ln(x) - x + C \tag{17}$$

$$\int \log_a(x) \ \mathrm{d}x = \frac{1}{\ln(a)}(x\ln(x) - x) + C \qquad (a > 0 \,\text{and}\, a \neq 1) \tag{18}$$

$$\int \sin(x) \ \mathrm{d}x = -\cos(x) + C \tag{19}$$

$$\int \cos(x) \ \mathrm{d}x = \sin(x) + C \tag{20}$$